

NOTRE SOLUTION

 Générateur d'écrans d'erreur factices 

Plateforme de simulation d'écrans OS

 15 écrans prédéfinis (Windows, Linux, macOS, Ransomware)

 Mode plein écran — touche **Q** pour quitter

Communauté

 URL partageable par écran

 Upload privé / Upload publique

RGPD

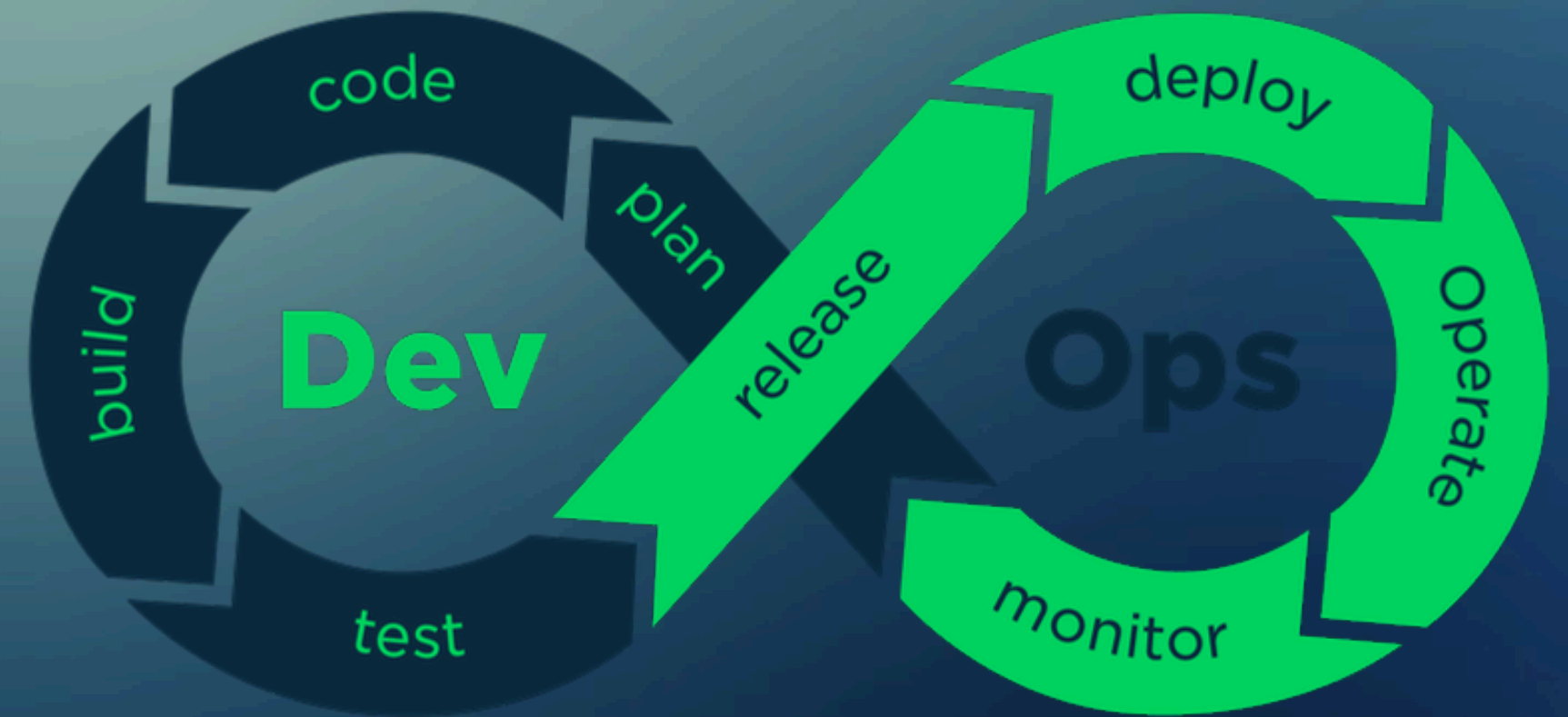
 Aucune données stockés

 Suppression instantanée

 Données Exif effacées

SHIFT LEFT

VERIFICATION / CHECK
AVANT DEPLOIEMENT



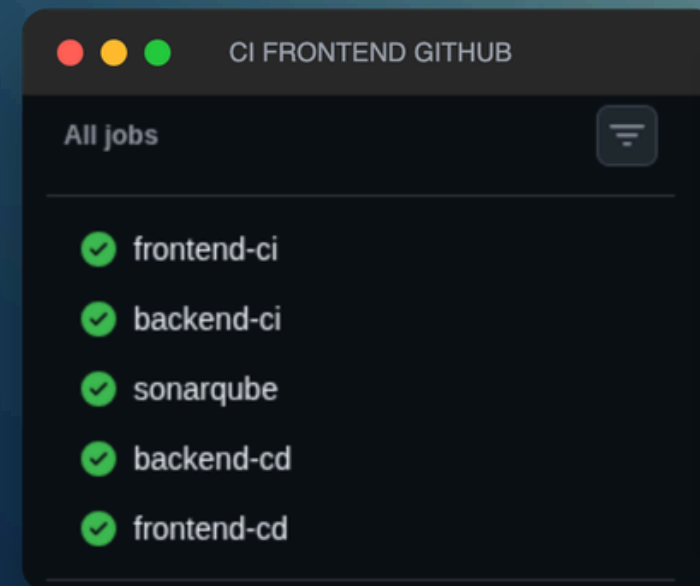
SHIFT LEFT

SHIFT RIGHT

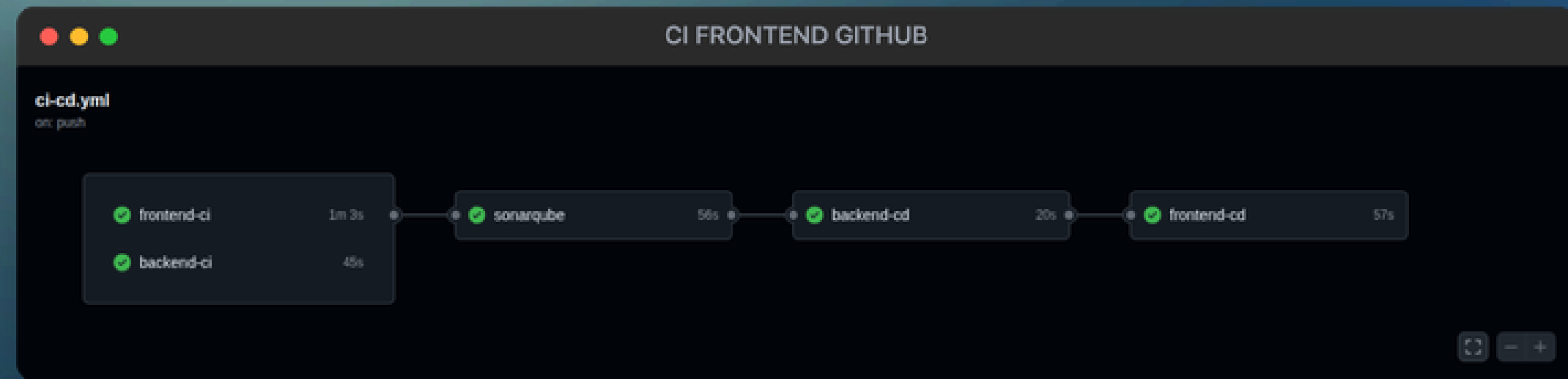
CI / CD


CI_CD.yaml

1 workflow



5 jobs

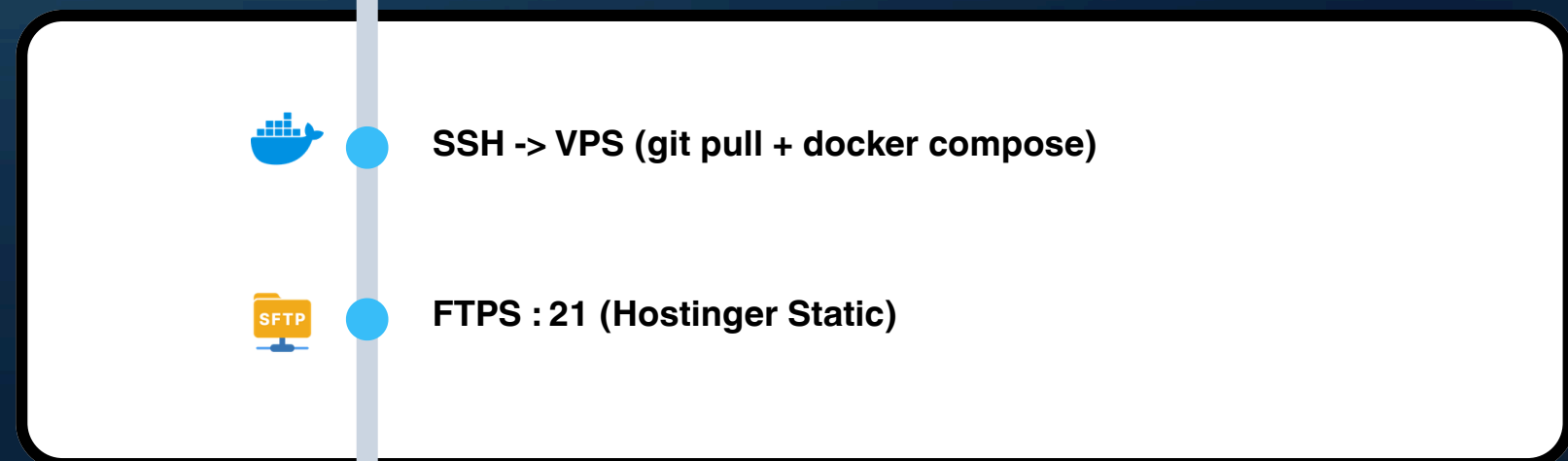
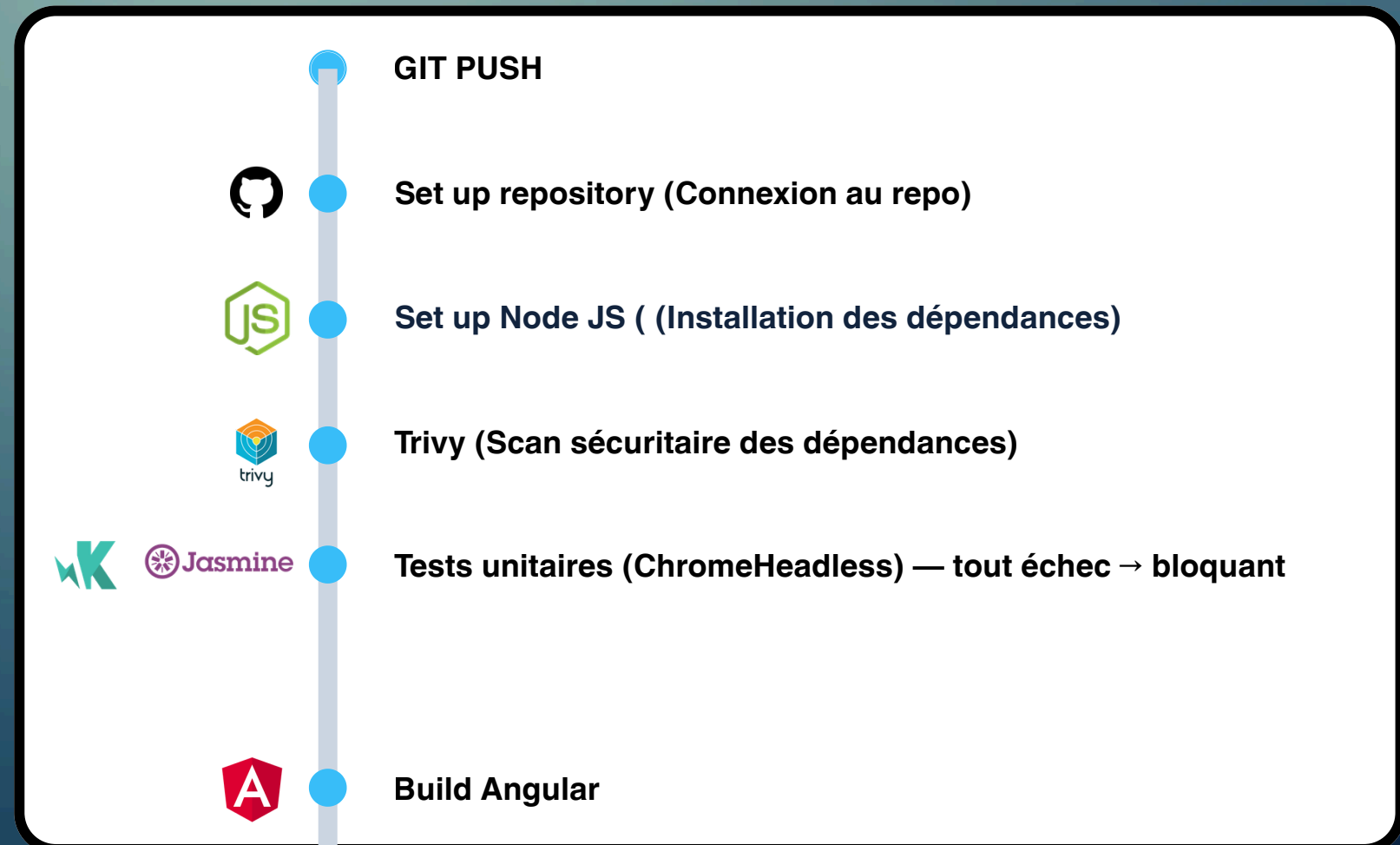


CD dépend de chaque CI

FRONT END



Schema CI Frontend

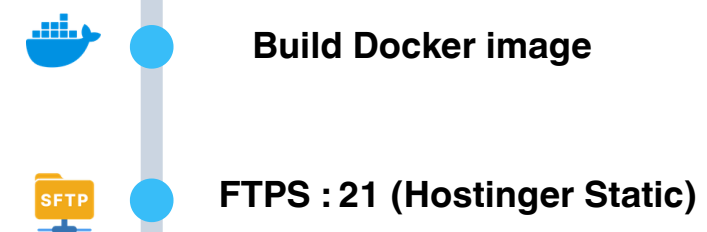
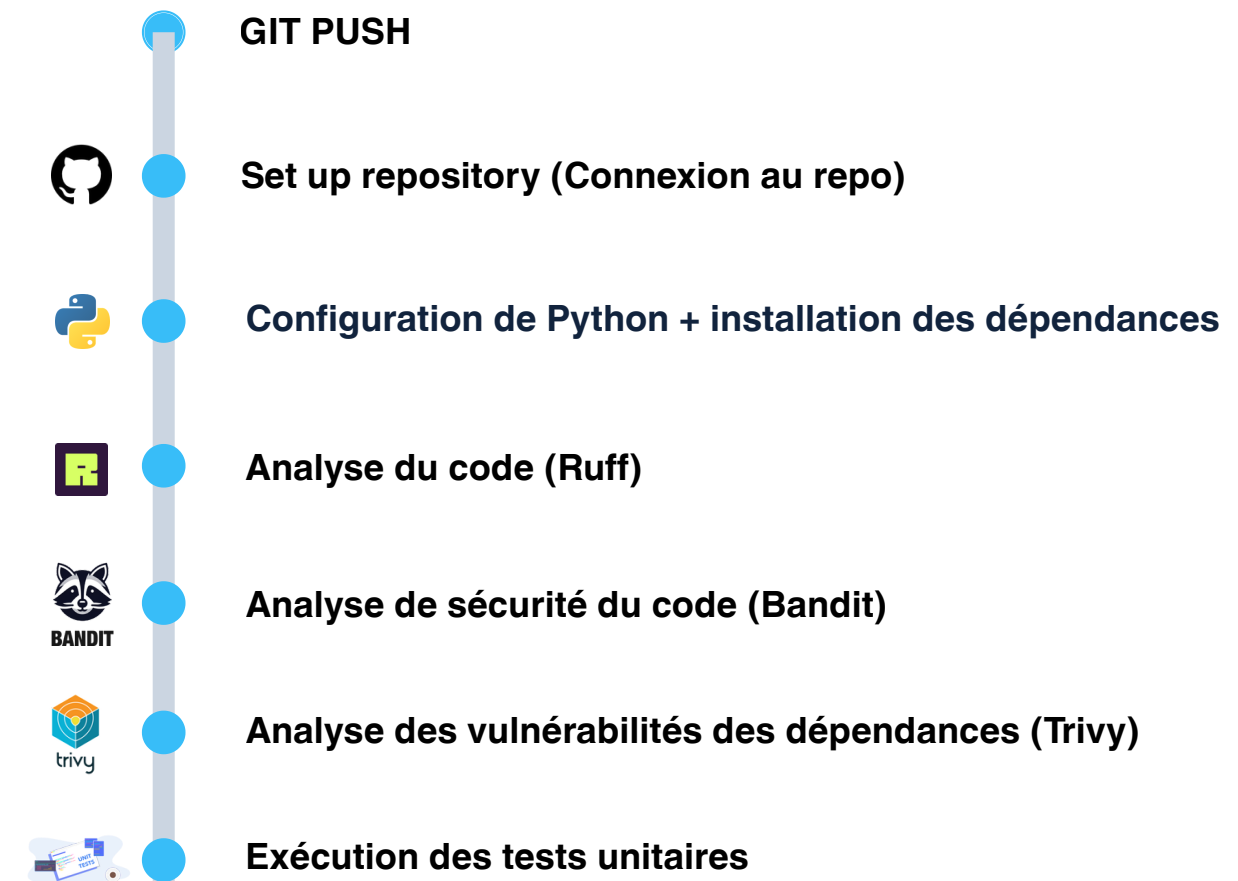


Schema CD Frontend

BACK END



Schema CI Backend



Schema CD Frontend

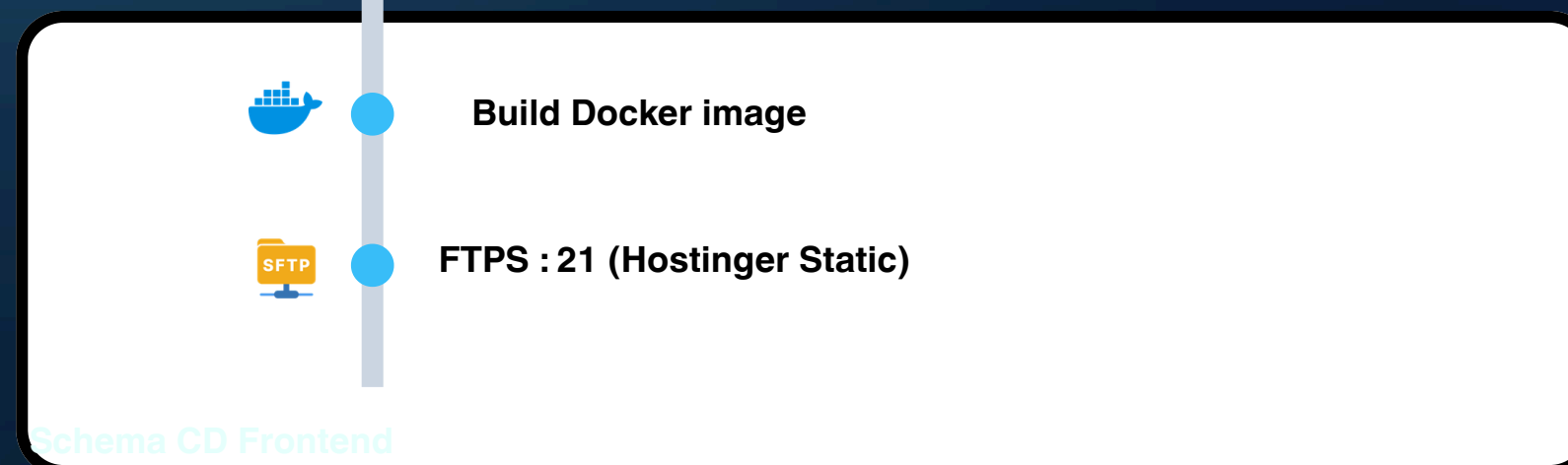
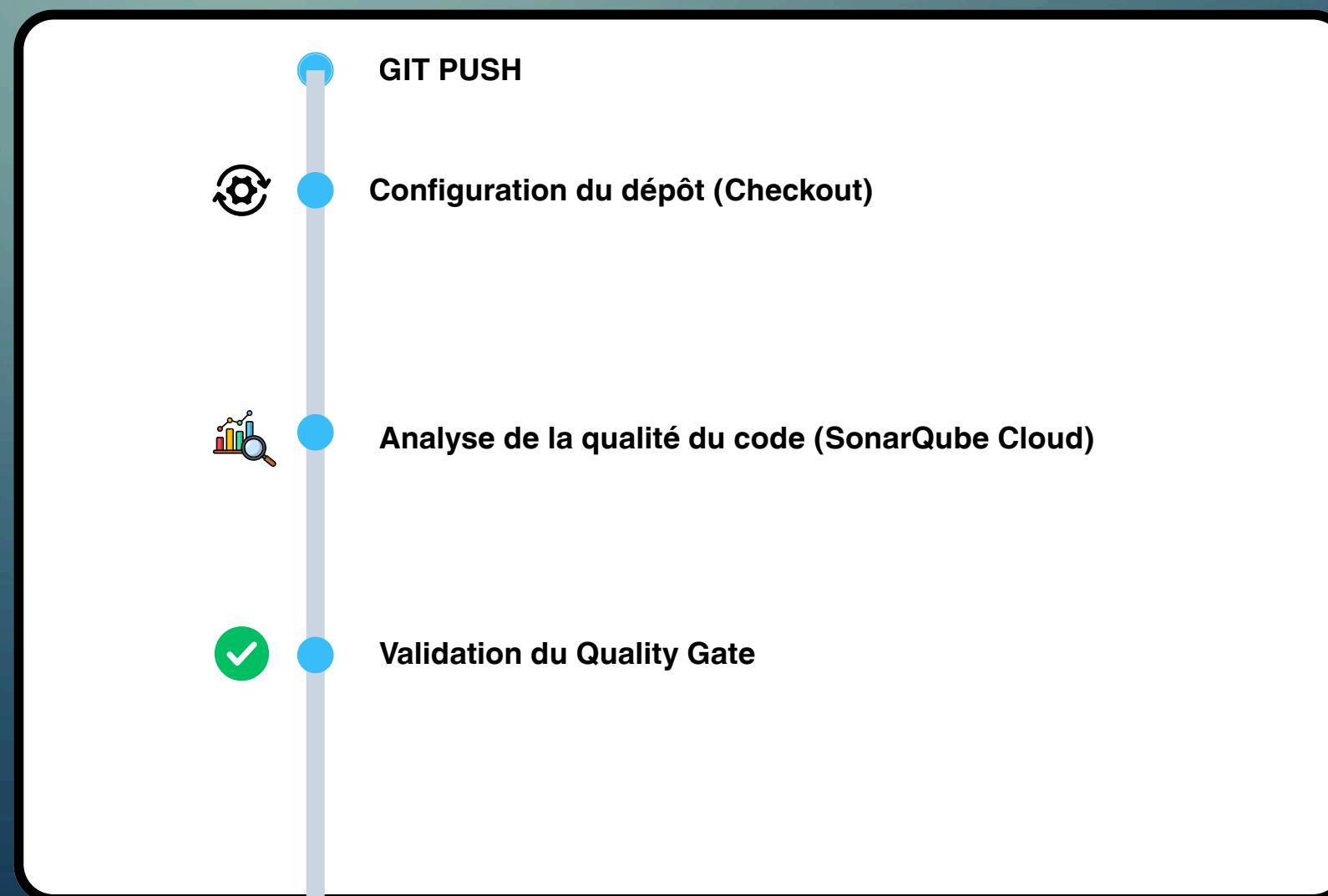
SONAR QUBE



A REVOIR

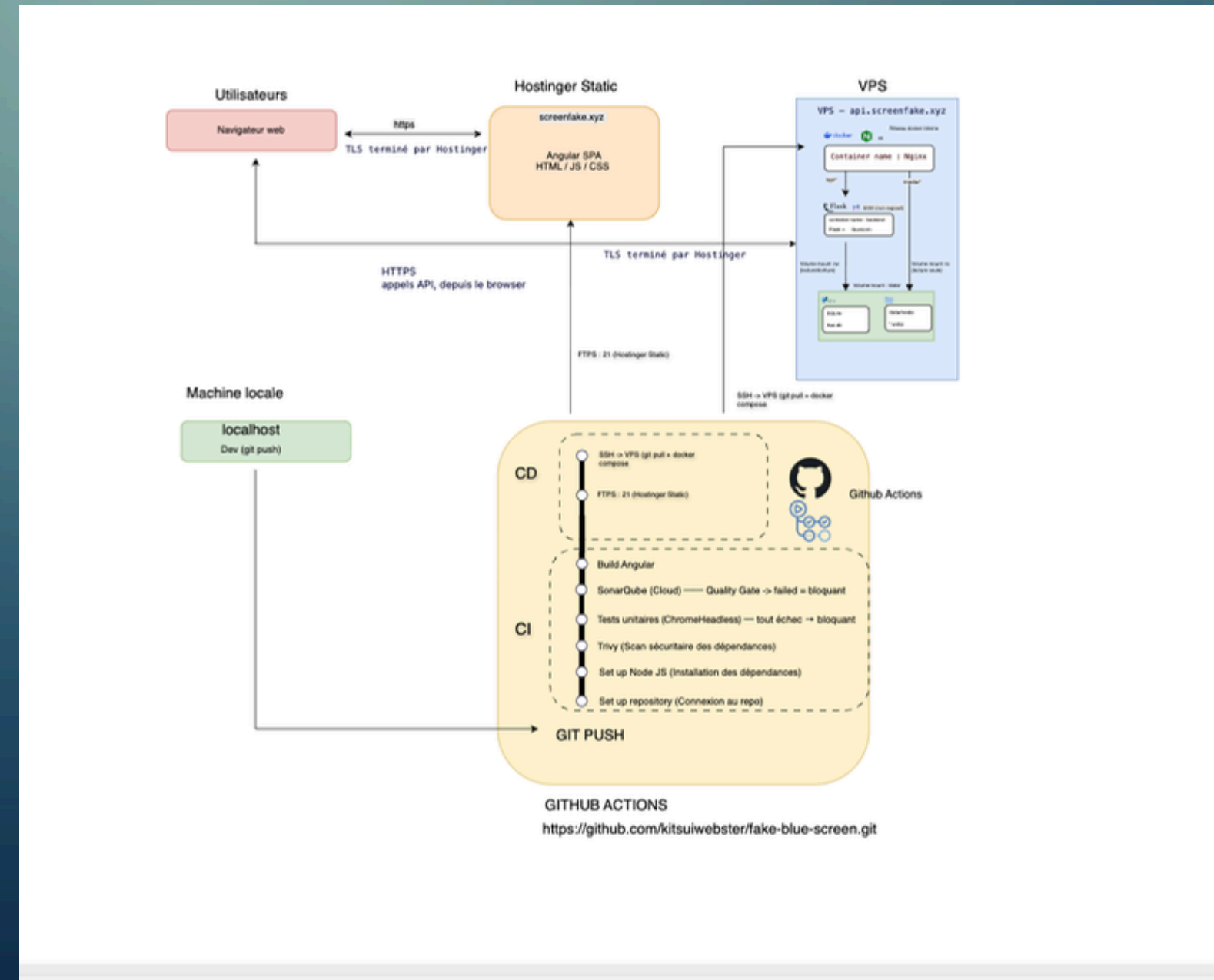


Schema CI Backend



Schema CD Frontend

SCHEMA ARCHI



Schema CD Frontend

GESTION DES PAQUET & SUPPLY CHAIN ≡

OUTIL

Trivy

Comment on l'utilise ?

- Intégré dans notre pipeline CI/CD pour détecter les vulnérabilités critiques
- Réduit risques d'exploitation via des bibliothèques compromises

Pertinence ?

Scanne les vulnérabilités du code source et les dépendances.

```
Trivy vulnerability scan

78 Report Summary
79
80
81 | Target | Type | Vulnerabilities |
82 |-----|-----|-----|
83 | requirements.txt | pip | 2 |
84 |-----|-----|-----|
85 Legend:
86 - '-': Not scanned
87 - '0': Clean (no security findings detected)
88
89
90 requirements.txt (pip)
91 =====
92 Total: 2 (HIGH: 2, CRITICAL: 0)
93
94 | Library | Vulnerability | Severity | Status | Installed Version | Fixed Version | Title |
95 |-----|-----|-----|-----|-----|-----|-----|
96 | Pillow | CVE-2025-48379 | HIGH | fixed | 11.2.1 | 11.3.0 | python-pillow: pillow: Pillow DOS Heap Buffer Overflow |
97 | | | | | | | https://avd.aquasec.com/nvd/cve-2025-48379 |
98 | | | | | | | |
99 | | | | | | | |
100 | | CVE-2026-25990 | | | | 12.1.1 | pillow: Pillow: Out-of-bounds Write via Specially Crafted |
101 | | | | | | | PSD Image |
102 | | | | | | | https://avd.aquasec.com/nvd/cve-2026-25990 |
103 |-----|-----|-----|-----|-----|-----|-----|
```


TESTS DE SECURITE AUTOMATISES

3 étapes clé



Jasmine
rôle - écrit les règles



Framework de Test



Fichier (spec.ts)



Définit 2 fonctions pour karma

L'application (SPA) se lance
correctement

L'onglet "Galerie" s'affiche bien



Karma
Rôle — Configure et envoie les tests



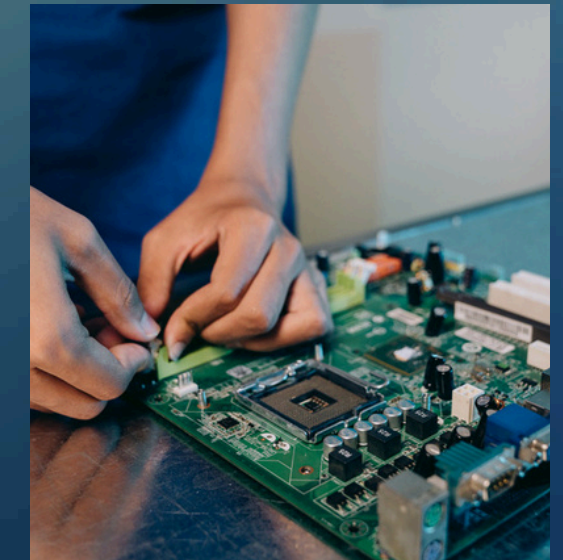
Traduit le TS → JS



Ouvre & Injecte le code dans
ChromeHeadless



Retourne résultat à Github
Action



Chrome Headless
Rôle — exécute dans un navigateur



Moteur Chrome headless



Exécute les test



Retourne le
résultat à Karma



IMPLEMENTATION

CI FRONTEND GITHUB

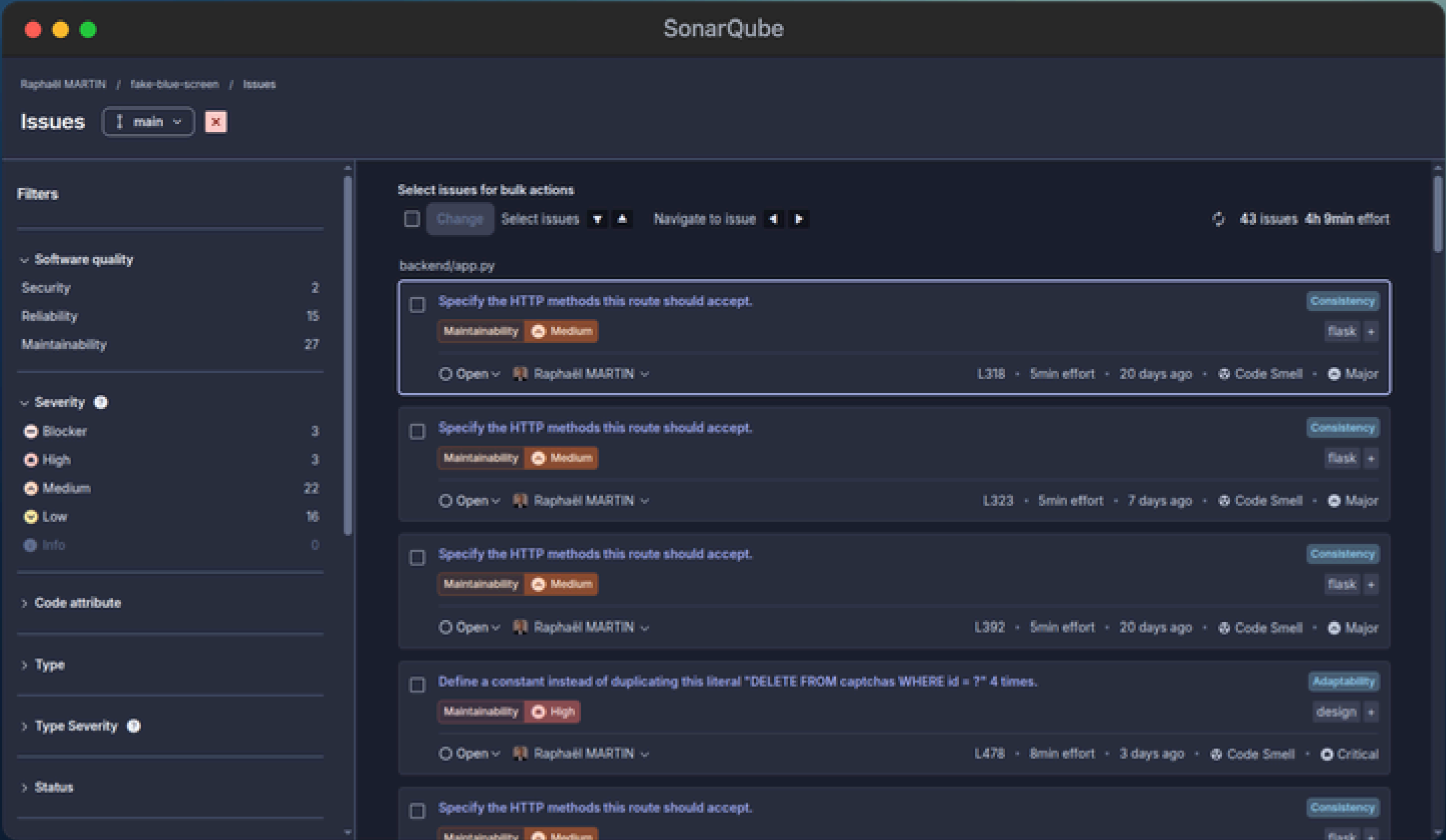
Unit tests (headless)13s

```
1 ▶ Run yarn test -- --watch=false --browsers=ChromeHeadless
16 yarn run v1.22.22
17 warning From Yarn 1.0 onwards, scripts don't require "--" for options to be forwarded. In a future version,
   any explicit "--" will be forwarded as-is to the scripts.
18 $ ng test --watch=false --browsers=ChromeHeadless
19 > Building...
20 ✓ Building...
21 Initial chunk files | Names | Raw size
22 chunk-JU77SJCG.js | - | 2.45 MB |
23 polyfills.js | polyfills | 1.04 MB |
24 spec-app-app.spec.js | spec-app-app.spec | 390.90 kB |
25 jasmine-cleanup-1.js | jasmine-cleanup-1 | 67.20 kB |
26 test_main.js | test_main | 59.92 kB |
27 styles.css | styles | 24.54 kB |
28 chunk-TTULUY32.js | - | 1.99 kB |
29 jasmine-cleanup-0.js | jasmine-cleanup-0 | 519 bytes |
30
31 | Initial total | 4.03 MB
32
33 Application bundle generation complete. [2.338 seconds] - 2026-03-11T18:59:33.807Z
34
35 11 03 2026 18:59:33.900:INFO [karma-server]: Karma v6.4.4 server started at http://localhost:9876/
36 11 03 2026 18:59:33.901:INFO [launcher]: Launching browsers ChromeHeadless with concurrency unlimited
37 11 03 2026 18:59:33.905:INFO [launcher]: Starting browser ChromeHeadless
38 11 03 2026 18:59:42.326:INFO [Chrome Headless 145.0.0.0 (Linux 0.0.0)]: Connected on socket
   R9wEaRWmbdCSj7tdAAAB with id 58557868
39 Chrome Headless 145.0.0.0 (Linux 0.0.0): Executed 0 of 2 SUCCESS (0 secs / 0 secs)
40 Chrome Headless 145.0.0.0 (Linux 0.0.0): Executed 1 of 2 SUCCESS (0 secs / 0.205 secs)
41 Chrome Headless 145.0.0.0 (Linux 0.0.0): Executed 2 of 2 SUCCESS (0 secs / 0.21 secs)
42 Chrome Headless 145.0.0.0 (Linux 0.0.0): Executed 2 of 2 SUCCESS (0.243 secs / 0.21 secs)
43 TOTAL: 2 SUCCESS
44 Done in 13.06s.
```

> Build7s



SONARQUBE



Rôle

Analyse statique du code (SAST)

Détection de bugs, vulnérabilités et code smells



Intégration dans la CI

Configuration via sonar-project.properties

Scan déclenché après les tests unitaires

Authentification via SONAR_TOKEN (secret GitHub)



Résultat

SonarQube calcule un Quality Gate

Échec du Quality Gate → problème signalé dans SonarCloud

GESTION DES SECRETS

3 étapes clé



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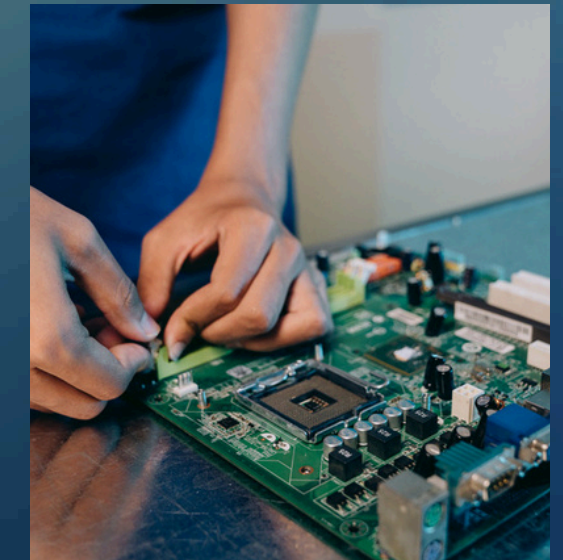
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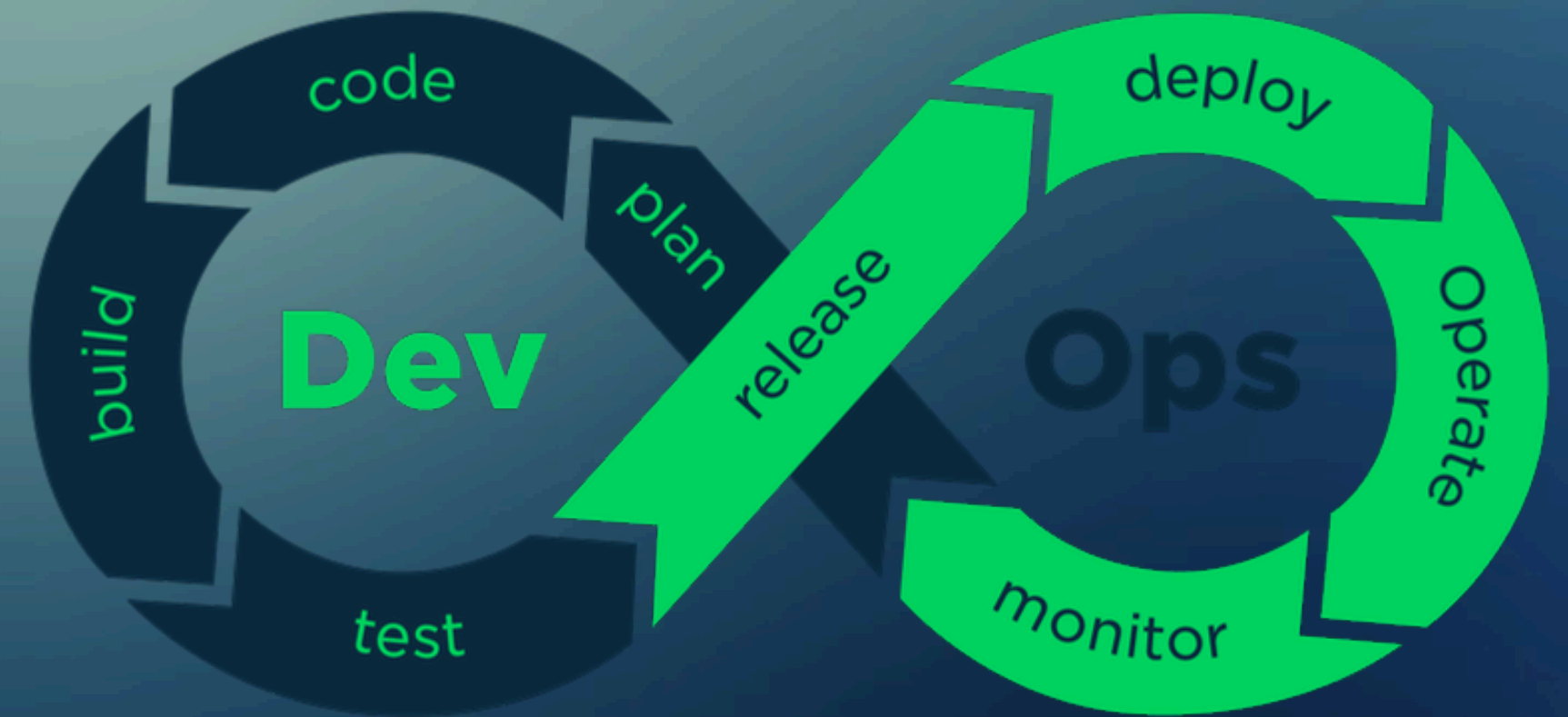
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Retourne le
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SHIFT RIGHT

VERIFICATION / CHECK
APRES DEPLOIEMENT



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MONITORING

CI FRONTEND GITHUB

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DÉTECTION & RÉPONSE

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DEMO



01

